

Chapter 3 — In the Earliest Cities

The Harappan / Indus Valley Civilisation

The one-line point: ~4,700 years ago, people on the Indus built **planned cities with grid streets, covered drains, standardised bricks and public baths** — and then, ~800 years later, it all ended and nobody knows exactly why. **This is the highest-yield chapter in the book.**

← Ch.2 | Index | Next: Ch.4 →

Syllabus Classification

Field	Value
Subject	History → Ancient India
Sub-discipline	Proto-history / Bronze Age Urbanism (proto-history = writing exists but is undeciphered)
Period	~2600 BCE – 1900 BCE (mature phase); ~4,700 to ~3,900 years ago
Prelims (GS Paper I)	<i>History of India</i> → the single most-asked ancient India topic
Mains (GS Paper I)	<i>Indian culture — art forms & architecture from ancient times</i> : town planning, seals, sculpture, drainage
Also feeds	Art & Culture (seals, Dancing Girl, Priest-King, Pashupati seal), Geography (river systems, Ghaggar-Hakra), Environment (water management, urban decline & climate), Governance (sanitation, urban planning as a model)
Weightage	★★★★★ — asked in roughly every alternate Prelims ; multiple Mains appearances
Overlaps with	Current affairs : Rakhigarhi DNA studies, Dholavira's UNESCO World Heritage inscription (2021), Aryan migration debate

 **Not Modern History. This is Ancient India → Proto-history**, and it crosses heavily into **Art & Culture**. If you learn one chapter from this book properly, make it this one.

Learning Outcomes

By the end of this chapter you should be able to:

- ☐ **Describe** the standard Harappan city layout — citadel vs lower town — and say what each contained.
- ☐ **Explain** the drainage system in technical detail and argue why it implies a central authority.
- ☐ **Locate and match** all major sites to their unique features: Mohenjodaro, Harappa, Lothal, Dholavira, Kalibangan, Chanhudaro, Rakhigarhi, Banawali, Surkotada, Nageshwar, Balakot, Shortughai.
- ☐ **List** the crops, animals and technologies, and identify what the Harappans **did not** have.
- ☐ **Explain** why the seals and script matter, and state precisely what we can and cannot conclude from an undeciphered script.
- ☐ **Trace** the raw-material trade network — which material came from where — and name the Mesopotamian trade terms (Meluhha, Magan, Dilmun).
- ☐ **Explain** the weight system and why standardisation is evidence of centralised control.

- ☐ **Evaluate** the competing theories for the civilisation's decline, without overcommitting to one.
- ☐ **Compare** Harappan and Vedic cultures point-by-point (this is a recurring Prelims format — see Ch.4).
- ☐ **Deploy** Harappan town planning in a Mains answer on ancient Indian urbanism or water management.

Start Here: What "First City" Actually Means

You've read about villages in Chapter 2 — a few hundred people, mud huts, everyone farming.

Now picture this instead:

A city of 35,000–40,000 people. Streets laid out on a grid, crossing at right angles. Every house connected to a covered drain running beneath the street, with inspection holes for cleaning. Bricks in a fixed 1:2:4 ratio, identical from Gujarat to Afghanistan — a distance of over 1,500 km. A public water tank made watertight with gypsum mortar and bitumen. Standardised stone weights in a binary sequence.

Now the part that should genuinely unsettle you: **this was 4,700 years ago.** Many Indian towns did not have covered sewerage in 1947.

WHO ELSE WAS BUILDING CITIES AT THIS MOMENT? (~2600 BCE)	
MESOPOTAMIA	– Sumer, Ur, Uruk (Tigris–Euphrates)
EGYPT	– Old Kingdom, the Great Pyramids
HARAPPA	– Indus valley
CHINA	– a bit later (Shang, ~1600 BCE)
THE HARAPPAN WAS THE LARGEST OF ALL OF THEM BY AREA – roughly 1.5 million sq km, ~1,500 km east-to-west	

And here's the twist that defines the whole chapter: Egypt gave us pyramids and pharaohs' names.

Mesopotamia gave us kings and epics. **Harappa gave us drains, weights and grids — and not one single ruler's name.** We cannot read a word they wrote.

The Software Engineer's Analogy

The Harappan civilisation is **a large-scale distributed system whose source code we possess but cannot decompile.**

WHAT WE HAVE	WHAT IT TELLS US
Identical brick ratio 1:2:4 across 1,500 km	→ A SHARED SPEC, centrally defined and enforced. Nobody achieves this by accident.
Standardised weights in 1,2,4,8,16,32... sequence	→ A STANDARDS BODY. Trade needs agreed units. Someone published and policed the standard.
Grid streets + covered drains laid BEFORE houses	→ CENTRAL PLANNING, and planning BEFORE building. Infrastructure first, then the app layer.
Seals with a script we cannot read	→ An ENCODED PROTOCOL we have packets for but no schema. ~4,000 samples, avg 5 signs each – too little data to crack.
NO palaces, NO grand tombs, NO royal names	→ Either a very different power structure, or one that left no monuments to itself. UNRESOLVED.

🔑 **The takeaway to carry into every Harappan question:** we have extraordinary evidence about their *engineering* and almost none about their *politics or religion*. **So be very careful about claims regarding Harappan rulers, gods or warfare** — that's exactly where Prelims sets its traps.

1 Discovery — How We Found Them

This is a genuinely good story, and the names are asked.

TIMELINE OF DISCOVERY

- 1850s → British engineers building the Lahore-Multan RAILWAY find a mound of high-quality ancient bricks at Harappa. They GRIND THEM UP and use them as track ballast. An unknown quantity of the city is gone forever.
- 1875 → Alexander Cunningham (first Director-General of the ASI) reports a seal from Harappa. He cannot place it, and wrongly assumes it is foreign.
- 1921 → DAYA RAM SAHNI excavates HARAPPA (Punjab, Pakistan)
- 1922 → R. D. BANERJI excavates MOHENJODARO (Sindh, Pakistan)
- 1924 → JOHN MARSHALL (ASI Director-General) announces to the world that a previously unknown civilisation has been found. India's history is instantly pushed back ~2,000 years.
- Post-1947 → Partition puts Harappa and Mohenjodaro in Pakistan. Indian archaeologists search for sites in India and find LOTHAL, KALIBANGAN, DHOLAVIRA, RAKHIGARHI, BANAWALI...
- 2014-2025 → RAKHIGARHI (Haryana) emerges as possibly the LARGEST Harappan site; ancient DNA studies from here become central to the Aryan migration debate.
- 2021 → DHOLAVIRA inscribed as a UNESCO WORLD HERITAGE SITE

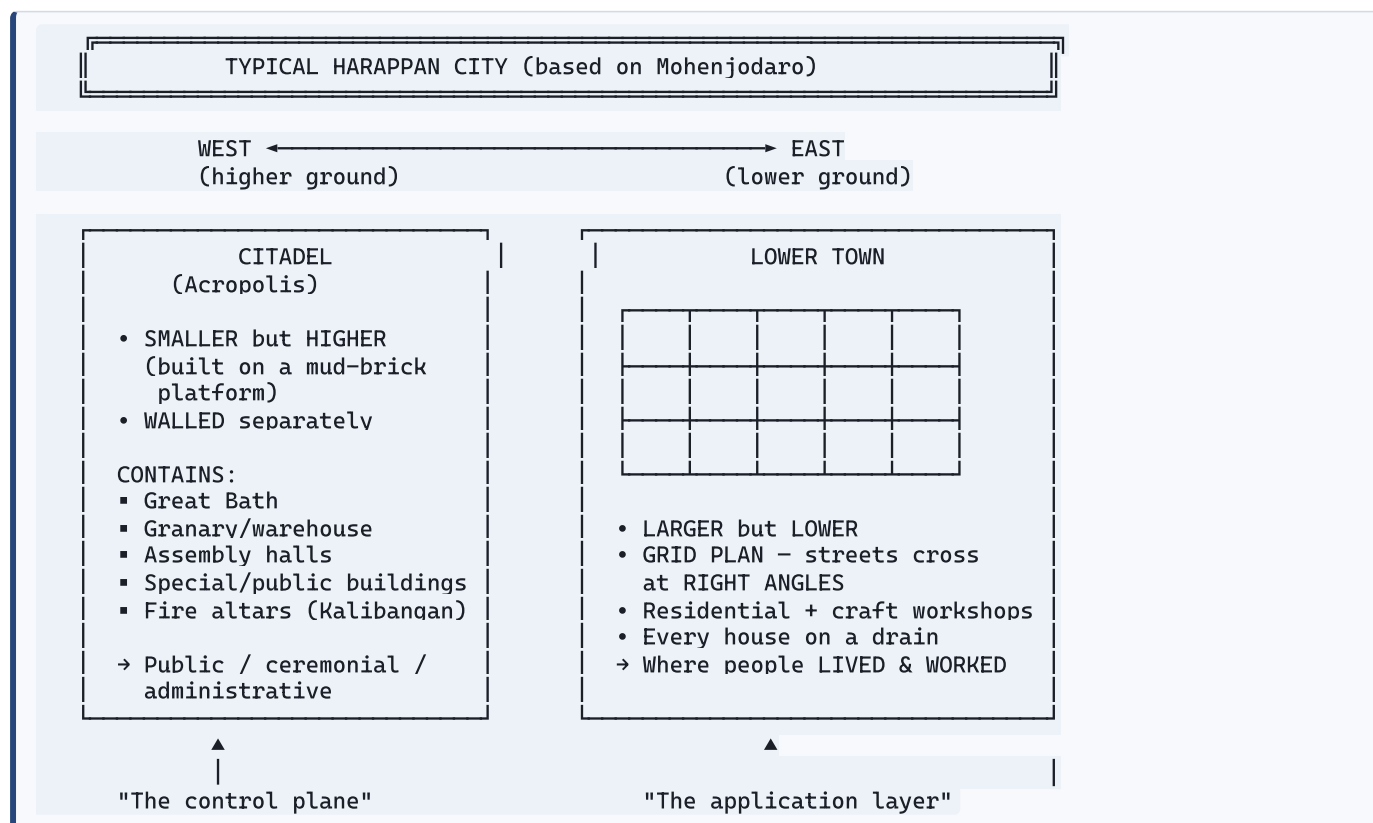
The naming question — "Harappan" or "Indus Valley"?

Name	Why used	Why it's imperfect
Harappan Civilisation	Archaeological convention: name a culture after the first site excavated — Harappa (1921)	Harappa wasn't the capital or the biggest site
Indus Valley Civilisation	Most early sites lie along the Indus and its tributaries	Misleading — many sites (Lothal, Dholavira, Kalibangan, Rakhigarhi) are <i>not</i> in the Indus valley. Over half lie along the dried-up Ghaggar-Hakra river
Indus-Saraswati Civilisation	Used by those identifying Ghaggar-Hakra with the Rigvedic Saraswati	Contested; the identification is debated

🎯 **Prelims-worthy nuance:** "Harappan Civilisation" is the **more accurate** term precisely because the civilisation extends well beyond the Indus valley. NCERT uses "Harappan."

2 The City Layout — The Master Diagram

Draw this from memory. It underpins half the questions on this chapter.



The rules of Harappan urban planning

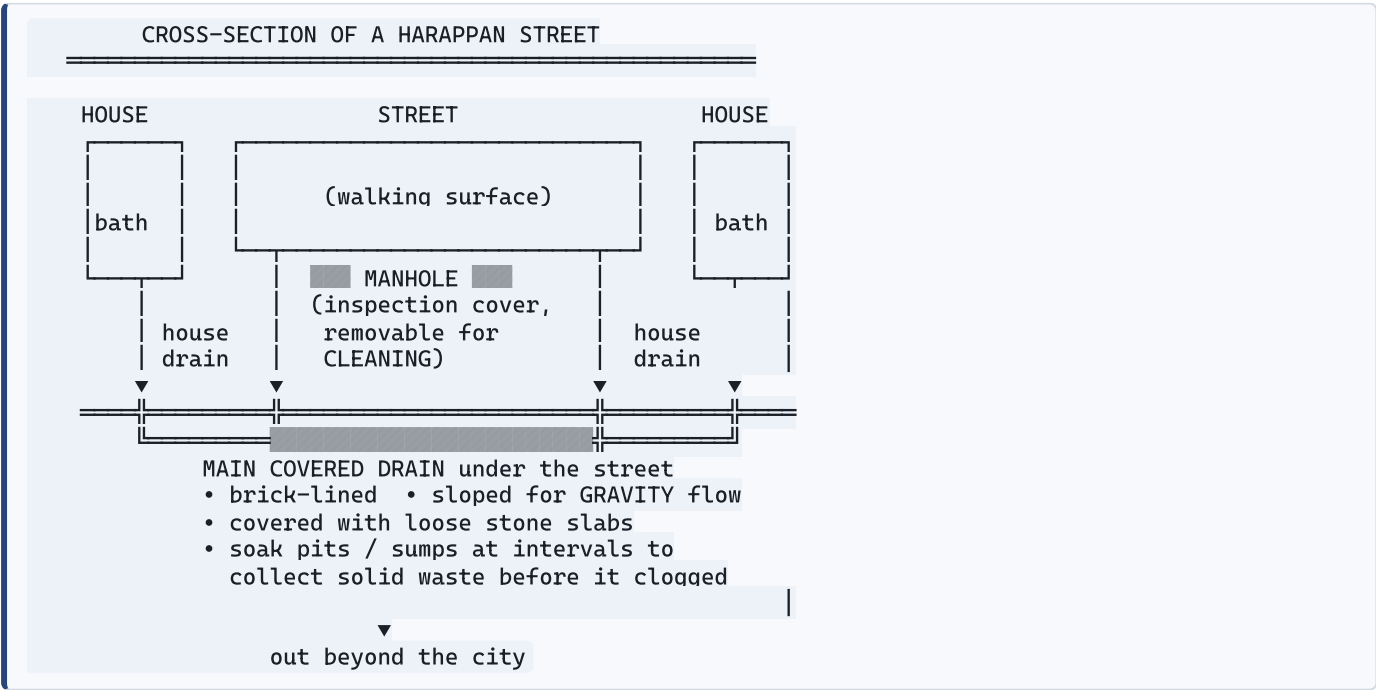
Feature	Detail
Two-part division	Citadel (west, smaller, higher, walled) + Lower Town (east, larger, lower, walled). Both usually walled
Grid street plan	Streets and lanes cross at right angles , dividing the city into rectangular blocks
Baked bricks	Standardised and remarkably well-made; laid in an interlocking pattern for strength
Brick ratio	1 : 2 : 4 (thickness : width : length) — the <i>same ratio</i> across the entire civilisation
Drains first	Drains were laid out BEFORE the houses were built — the killer detail
Wells	Many houses had their own well ; Mohenjodaro had ~700 wells
No temples or palaces	No structure has been <i>conclusively</i> identified as either

★ **"Drains were planned before houses" is the single most quotable line in this chapter.** Think about what it requires: someone surveyed the site, designed a network, dug and built it, and only then allowed construction — and everyone complied. **That is a state, or something very like one.**

🖥️ You'd call it *infrastructure-as-code, provisioned before deployment*. You cannot retrofit a gradient-correct sewer network under an existing city. It must be designed first. So must a civilisation.

3 The Drainage System — Their Greatest Achievement

Take this seriously. It is what makes the Harappans exceptional.



Why this is genuinely remarkable

Feature	What it required
Every house connected	Universal coverage — not a privilege of the rich. Suggests a <i>civic</i> system, not an elite one
Covered drains	Understanding that open sewage causes disease and smell
Gradient/slope	Careful surveying — water must flow by gravity across the whole city
Manholes / inspection holes	Designed for maintenance , not just construction. Someone was responsible for ongoing cleaning
Soak pits / sumps	Solid waste settled out before it could block the line
Standard brick sizes	Manufacturing standards, and a supply chain to enforce them

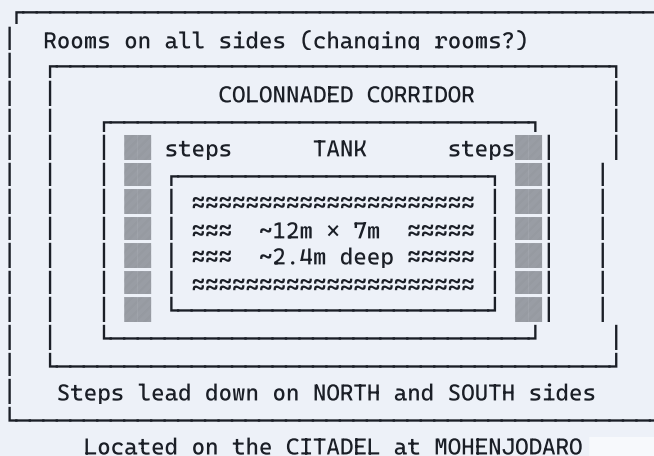
 **They designed for observability and maintenance.** Manholes are access ports; soak pits are a filtering stage that prevents downstream failure. This isn't just building a thing — it's building a thing that can be *operated* for generations. Most ancient engineering doesn't do this.

The organisational implication (say this in a Mains answer): a system like this cannot be built or maintained by individual households acting alone. It requires **surveying, standardisation, coordinated construction, and continuous municipal maintenance** — in other words, a functioning **civic authority** with the power to plan and enforce.

4 The Star Structures

The Great Bath (Mohenjodaro)

The most famous Harappan structure.



Detail	Value
Location	Citadel , Mohenjodaro
Size	~12 m long × 7 m wide × 2.4 m deep
Waterproofing	Bricks set on edge with gypsum mortar ; a layer of bitumen (natural tar) lining the sides and floor
Water supply	From a well in an adjacent room
Drainage	A large corbelled drain in one corner emptied the tank
Surroundings	Corridor on all four sides; rooms on three sides — one with a well
Purpose	Probably ritual bathing on special occasions — a public/ceremonial function

 **The Great Bath is your evidence for "ritual purity mattered."** But be careful: we're *inferring* ritual from architecture. It's the leading hypothesis, not a proven fact — and Prelims respects that distinction.

 **Why it's astonishing engineering:** they solved waterproofing 4,700 years ago using gypsum mortar plus bitumen — a two-layer system. That's a genuine materials-science achievement.

The Granary / Warehouse

Found at **Mohenjodaro** and **Harappa** (Harappa's is near the river, with circular brick platforms nearby, possibly for threshing grain).

- Large structures on the **citadel**, with **ventilation ducts** and raised platforms (keeps grain dry, deters rodents)
-  **Terminology caution:** these were identified as "granaries" by early excavators, but the identification is **disputed** — no significant grain residue was found in Mohenjodaro's. Many archaeologists now call them "warehouses" or "great halls." NCERT still uses "granary."

 **Why a granary on the citadel matters:** collecting and storing grain centrally implies **taxation or tribute** — someone gathered surplus from farmers and stored it under guard. This is the direct continuation of Chapter 2's "storage → surplus → the state" chain.

Fire Altars

Found at **Kalibangan** (Rajasthan) and **Lothal** (Gujarat).

- Rows of raised brick platforms with ash, charcoal and terracotta cakes
- At Kalibangan, found both on the citadel *and* inside private houses
- Interpreted as evidence of **fire rituals / sacrifice**

 **Why this is significant for a bigger debate:** fire ritual is central to *Vedic* religion (Chapter 4). Fire altars at Harappan sites are used by some scholars to argue for continuity between Harappan and Vedic culture. **Contested, but very much asked.**

The Dockyard (Lothal) — ★★ ★

Lothal, Gujarat — one of the most important sites for India.

Feature	Detail
The dockyard	A large brick-lined basin , ~214 m × 36 m, connected to the Bhogavo river/Gulf of Khambhat by a channel
What it did	Ships and boats came in; a sluice gate and spill channel maintained water level at low tide
Why it matters	The world's earliest known dockyard (as identified). Lothal was a port and trading hub
Also at Lothal	A bead-making factory (with kiln, raw material, finished and unfinished beads), a warehouse , fire altars , a Persian Gulf seal, and rice cultivation
Layout note	Unlike most sites, Lothal's citadel and lower town were enclosed within one wall

 **Lothal = trade + beads + dockyard + rice + fire altar.** Four or five separate "famous for" tags on one site. Very high frequency in Prelims.

Dholavira (Gujarat, Rann of Kutch) — ★★

UNESCO World Heritage Site (2021) — which makes it current-affairs-relevant too.

Feature	Detail
Unique layout	Divided into THREE parts (not two): Citadel, Middle Town, Lower Town — each with massive stone walls
The signboard ★	A large inscription of ~ 10 large Harappan signs , made of white gypsum inlay on wood, found near the citadel gate — the world's earliest known "signboard"
Water management ★	An extraordinary system of stone-cut reservoirs , check dams and channels to harvest every drop of monsoon runoff — Dholavira sits in an arid zone between two seasonal streams (Mansar and Manhar)
Building material	STONE masonry (most Harappan sites are brick) — local stone was abundant
A "stadium"	A large open ground with terraced seating — possibly for ceremonies or public gatherings

 **Modern relevance:** Dholavira is regularly cited in water-conservation and rainwater-harvesting discussions. **Use it in Mains/Essay answers on water security** — an ancient Indian city in a desert that survived on engineered rainwater harvesting. That's a strong, specific example.

5 The Site Map — Learn Every One of These

HARAPPAN CIVILISATION – EXTENT & KEY SITES

AFGHANISTAN

- * SHORTUGHAI ← northernmost; LAPIS LAZULI trading outpost
- * MUNDIGAK

* MANDA (J&K) ← northernmost in India

- * HARAPPA (Ravi, Punjab-PAK)
Daya Ram Sahni, 1921
granary, cemetery R-37
- * RAKHIGARHI (Harvana)
LARGEST site in India
on Ghaggar; DNA studies

~~~~~ GHAGGAR-HAKRA (dried up) ~~~~~  
 ~~~ INDUS ~~~  
 * KALIBANGAN (Raj.)
 "black bangles"; PLOUGHED FIELD;
 FIRE ALTARS

- * MOHENJODARO (Sindh)
R.D. Banerji, 1922
GREAT BATH, granary,
Dancing Girl, Priest-King
- * BANAWALI (Haryana)
terracotta PLOUGH; barley

- * CHANHUDARO (Sindh)
ONLY city with NO CITADEL;
bead-making, shell, bangle

- * SUTKAGENDOR ← westernmost (Makran coast, PAK)

- * DHOLAVIRA (Kutch) — THREE-part city; SIGNBOARD;
water reservoirs; STONE; UNESCO 2021

- * LOTHAL (Gujarat) — DOCKYARD; beads; fire altar; RICE
- * SURKOTADA (Gujarat) — disputed HORSE bones
- * NAGESHWAR & BALAKOT — SHELL working
- * RANGPUR, ROJDI — rice, millets

- * DAIMABAD (Maharashtra) ← SOUTHERNMOST site
famous bronze chariot/animal figures

- * ALAMGIRPUR (UP) ← EASTERNMOST site

📖 The extremities — a classic Prelims question format

| Direction | Site | Location |
|--------------------------------|-------------|--------------------------------------|
| Northernmost | Manda | Jammu & Kashmir (on the Chenab) |
| Northernmost overall (outpost) | Shortughai | Afghanistan (lapis lazuli) |
| Southernmost | Daimabad | Maharashtra (on the Pravara) |
| Easternmost | Alamgirpur | Uttar Pradesh (on the Hindon) |
| Westernmost | Sutkagendor | Makran coast, Baluchistan (Pakistan) |

🔴 **Memory hook: "MDAS"** going N-S-E-W → **M**anda (N), **D**aimabad (S), **A**lamgirpur (E), **S**utkagendor (W).

The complete site table — your revision anchor

| Site | Location | River | Excavated by | ★ Famous for |
|----------------------------------|-----------------------------|-------------------------|-----------------------------|---|
| Harappa | Punjab, Pakistan | Ravi | Daya Ram Sahni, 1921 | First excavated; granary; cemetery R-37 ; workmen's quarters; bricks looted for railway |
| Mohenjodaro
★★★ | Sindh, Pakistan | Indus | R. D. Banerji, 1922 | GREAT BATH ; granary; Dancing Girl (bronze); Priest-King (steatite); Pashupati seal ; ~700 wells. Name = " <i>Mound of the Dead</i> " in Sindhi |
| Lothal ★★★ | Gujarat | Bhogavo | S. R. Rao | DOCKYARD ; bead factory; warehouse; fire altars ; rice ; Persian Gulf seal |
| Dholavira
★★★ | Kutch, Gujarat | Between Mansar & Manhar | J. P. Joshi / R. S. Bisht | THREE-part city ; SIGNBOARD ; water reservoirs; stone architecture; "stadium"; UNESCO 2021 |
| Kalibangan
★★★ | Rajasthan | Ghaggar | B. B. Lal / B. K. Thapar | "Black bangles" ; PLOUGHED FIELD (earliest known); FIRE ALTARS ; wooden furrow evidence |
| Chanhudaro
★★ | Sindh, Pakistan | Indus | N. G. Majumdar | The ONLY major city WITHOUT a citadel ; bead-making, shell and bangle workshops; a "craft town" |
| Rakhigarhi ★★ | Haryana | Ghaggar | Amarendra Nath | Largest Harappan site in India ; granary, cemetery; ancient DNA studies central to the Aryan debate |
| Banawali ★ | Haryana | Ghaggar | R. S. Bisht | Terracotta model of a PLOUGH ; good quality barley; both citadel and lower town but <i>no grid</i> streets |
| Surkotada ★ | Gujarat | — | J. P. Joshi | HORSE bones (heavily disputed); stone-covered burial |
| Nageshwar & Balakot ★ | Gujarat / Baluchistan coast | — | — | SHELL working — bangles, ladles, inlay |
| Shortughai ★ | Afghanistan | Oxus/Amu Darya | — | Trading outpost for LAPIS LAZULI |
| Sutkagendor | Makran coast, Pak | — | — | Westernmost ; port on the Iran route |
| Alamgirpur | Uttar Pradesh | Hindon | — | Easternmost |
| Daimabad | Maharashtra | Pravara | — | Southernmost ; famous bronze chariot and animal figurines |
| Manda | Jammu | Chenab | — | Northernmost in India |
| Ropar / Rupnagar | Punjab, India | Sutlej | Y. D. Sharma | First site excavated in independent India (1953); dog buried with human |

🧠 Memory hooks:

- **Kalibangan** → *kali* = **black** → "black bangles"; and it has the **ploughed field** ("Kalibangan — the farmer **banged** the plough into the field")
- **Chanhudaro** → "**Chanhu** has **no** citadel" (the odd one out)
- **Lothal** → "**Lot** of trade" → dockyard
- **Dholavira** → three parts, and its **signboard** — "Dhola-**VEER**-a" the *bold* one with the big sign
- **Surkotada** → "**Surkotada** = **horse**" (disputed)

6 Life, Craft and Technology

Agriculture

| Crops grown | Notes |
|----------------------------------|--|
| Wheat, barley | The staples |
| Pulses (lentil, chickpea) | Widespread |
| Sesame, mustard | Oil seeds |
| Rice | At Lothal and Rangpur (Gujarat) — relatively rare |
| Cotton ★ | The Harappans were among the world's first cotton growers . Greeks later called it " <i>Sindon</i> " — from Sindh |
| Millets | In Gujarat sites |

Farming technology:

- **The plough** — evidence from two sites:
 - **Kalibangan**: an actual **ploughed field** with two sets of furrows **crossing at right angles**, showing two crops grown together ★
 - **Banawali**: a **terracotta model of a plough** ★
- Ploughs were of **wood**, so the ploughs themselves haven't survived — only the field marks and the model
- **Irrigation** was needed (the region is semi-arid): canals found at **Shortughai**, and water reservoirs at **Dholavira**

Animals reared: cattle (very important), sheep, goat, buffalo, pig. Also **dogs** and **cats** (footprints of both, and a famous brick with a dog chasing a cat).

 **THE HORSE QUESTION — know this precisely.** Horse remains at Harappan sites are **rare and disputed** (Surkotada is the main claim, and specialists disagree on whether the bones are horse or wild ass/onager). There is **no depiction of a horse on any Harappan seal**. By contrast, the horse is *central* to Rigvedic society (Ch.4). **UPSC has asked this (2013)**. The safe formulation: "*Harappans did not use horse-drawn chariots in warfare; evidence for the horse is scanty and contested.*"

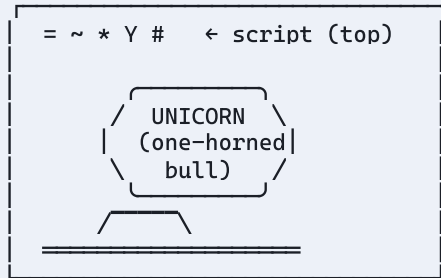
Crafts and materials

| Craft | Detail |
|----------------------|--|
| Bead-making ★ | Carnelian (a beautiful red stone), agate, jasper, crystal, quartz, steatite, faience, shell, terracotta, gold. Chanhudaro and Lothal were bead factories |
| Faience ★ | An artificial (man-made) material — ground sand/quartz mixed with colour and gum, shaped and fired to a glaze. Used for beads, bangles, small vessels. The Harappans manufactured a material that does not occur in nature |
| Shell working | Nageshwar and Balakot — bangles, ladles, inlay |
| Metallurgy | Copper and bronze (copper + tin). Also gold and silver . ❌ NO IRON — iron had not yet been discovered |
| Pottery | Fine wheel-made red pottery with black designs — plants, animals, geometric patterns |
| Textiles | Cotton and wool ; actual fragments of cotton cloth found; spindle whorls common |
| Seal-making | Steatite (soapstone), carved and then fired to harden |
| Stone tools | Chert blades, from the Rohri hills in Sindh |

⚠️ **NO IRON.** This is a Bronze/Chalcolithic civilisation. Any statement claiming Harappan iron tools is **false**. Iron enters India only around 1000 BCE (Chapter 4/5). Easy Prelims elimination.

◆ Seals and the Script

The **seal** is the iconic Harappan artefact. ~**4,000 inscribed objects** have been found.



Typical seal:

- STEATITE (soapstone)
- Usually SQUARE, ~2–4 cm
- Animal motif in the centre
- Script above (avg 5 signs)
- Boss/knob on the back with a hole → worn or tied on
- Pressed into wet clay to leave a SEALING

Animals shown: the **one-horned "unicorn"** (most common), **humped bull**, elephant, rhinoceros, tiger, buffalo, goat, and composite/mythical creatures. **Notably absent:** the **horse**, and the **lion**.

| Aspect of the script | What we know |
|------------------------|--|
| Number of signs | Roughly 375–400 distinct signs |
| Type | Too many signs for an alphabet, too few for a purely logographic system → probably logo-syllabic |
| Direction | Generally right to left . Longer texts are sometimes boustrophedon — right-to-left on one line, left-to-right on the next (like an ox ploughing a field) |
| Length | Very short — average ~5 signs , longest ~26 |
| Status | ⚠️ UNDECIPHERED |

Why can't we read it?

1. **No bilingual text** — nothing like the Rosetta Stone, which cracked Egyptian hieroglyphs by pairing them with Greek
2. **Texts are far too short** — 5 signs gives cryptanalysis almost nothing to work with
3. **We don't know the underlying language** — was it Dravidian? Indo-Aryan? Something extinct? Each assumption yields a different (unverifiable) decipherment
4. Even *whether it encodes language at all* is disputed by a minority of scholars

💻 **Precisely a cryptanalysis problem with a fatal data shortage.** You have ~4,000 ciphertext samples averaging 5 tokens each, no plaintext, no crib, and an unknown plaintext language. Frequency analysis needs volume and a known language model — you have neither. **The bottleneck isn't cleverness; it's data.** That's why 100 years of very smart people have failed.

What seals were probably FOR: stamping bundles of trade goods. Press the seal into wet clay on a package or a rope-tie; the sealing hardens; if it arrives intact, the goods weren't tampered with, and the mark identifies the sender.

💻 **A seal is a tamper-evident signature.** It doesn't encrypt; it *authenticates* and proves integrity in transit. Harappan sealings found in **Mesopotamia** are therefore direct evidence of goods travelling that far.

🏺 Weights and Measures

- Made of **chert** — a hard, uniform stone
- **Cubical**, mostly unmarked
- The sequence is **binary in the lower denominations: 1, 2, 4, 8, 16, 32, 64...** then shifts to decimal multiples of 16 (160, 200, 320, 640)
- **16 was clearly a key unit** ★ (a fun continuity: India used 16 *annas* to the rupee until 1957)
- The same weights turn up **across the entire civilisation**, from Gujarat to Afghanistan

📖 **A binary weight sequence isn't an accident.** With weights 1, 2, 4, 8, 16 you can compose *any* integer from 1 to 31 — the minimum set of weights for maximum coverage. That is a genuinely optimal design. Someone thought hard about this.

And the historical significance: standardised weights across 1.5 million sq km means **regulated, trusted, long-distance trade** — and an authority able to define and enforce a standard.

🎨 Famous artefacts (Art & Culture — know these by name and material)

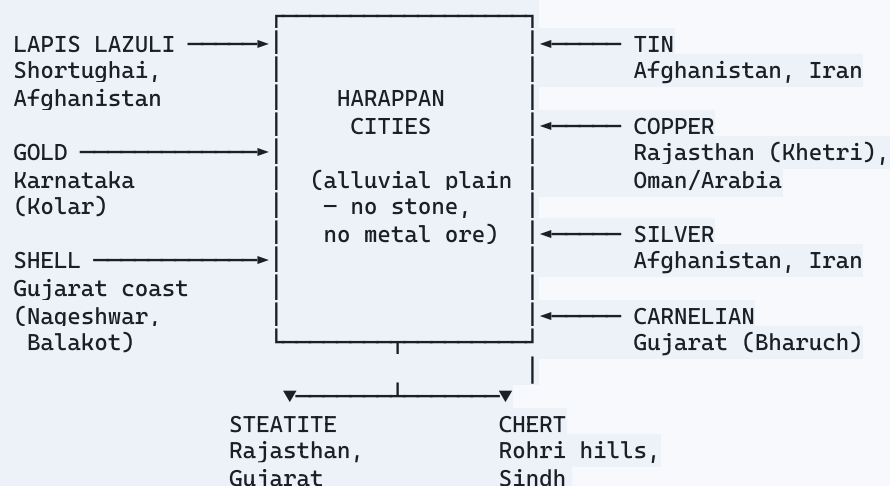
| Artefact | Site | Material | Note |
|--|-----------------|---|---|
| Dancing Girl
★★★ | Mohenjodaro | Bronze (lost-wax / <i>cire perdue</i> casting) | ~10.5 cm; a girl standing with hand on hip, bangles on one arm |
| Priest-King / Bearded Man
★★ | Mohenjodaro | Steatite | Trefoil-patterned shawl over the left shoulder; half-closed eyes |
| Pashupati Seal
★★★ | Mohenjodaro | Steatite | A seated yogic figure with a horned headdress, surrounded by elephant, tiger, rhino, buffalo, and deer beneath. Interpreted (contestably) as a proto-Shiva |
| Male Torso | Harappa | Red sandstone | Remarkably naturalistic |
| Bronze chariot & animals | Daimabad | Bronze | Famous hoard — chariot with rider, bull, elephant, rhinoceros |
| Mother Goddess figurines | Many sites | Terracotta | Numerous female figurines; often read as a fertility/mother-goddess cult |
| Terracotta toys | Many | Terracotta | Carts with movable wheels, whistles, animals, marbles — children existed ☺ |

🎯 **Learn the material, not just the name.** "Dancing Girl — bronze — Mohenjodaro" is a classic three-part matching question. So is "Priest-King — steatite" and "Male Torso — red sandstone."

7 Trade and Raw Materials — India's First Supply Chain

The core insight: the cities sat on flat alluvial plains that produce **grain but almost no stone, metal or gemstone**. Every hard material had to be imported. So the Harappans ran a continent-scale procurement network.

WHERE DID THE RAW MATERIALS COME FROM?



| Material | Source |
|----------------|---|
| Copper | Rajasthan (Khetri mines); also Oman/Arabia |
| Tin ★ | Afghanistan and Iran (rare — and essential to make bronze) |
| Gold | Karnataka (Kolar) |
| Silver | Afghanistan, Iran |
| Lapis lazuli ★ | Afghanistan (which is exactly <i>why</i> Shortughai exists) |
| Carnelian | Gujarat (Bharuch region) |
| Steatite | Rajasthan, Gujarat |
| Shell | Gujarat coast — Nageshwar, Balakot |
| Chert (blades) | Rohri hills, Sindh |
| Timber | Himalayan foothills, Gujarat |

🌐 International trade — the Mesopotamia connection

Mesopotamian texts (which we *can* read) mention trade with three lands. **This is where a readable civilisation tells us about an unreadable one.**

| Mesopotamian name | Almost certainly refers to |
|-------------------|--|
| MELUHHA ★★ ★ | The Harappan region (the Indus) |
| MAGAN | Oman |
| DILMUN | Bahrain (the intermediary island entrepôt) |

Goods Meluhha sent to Mesopotamia: carnelian beads, ivory, copper, gold, shell, wood, and — the poetic one — peacocks and monkeys.

The physical proof:

- **Harappan seals and sealings found in Mesopotamia** (Ur, Kish, Susa)
- **Etched carnelian beads** — a distinctly Harappan technique — found in Mesopotamian royal graves
- **Harappan-style cubical weights** found abroad
- A **Persian Gulf seal** (round, Dilmun type) found at **Lothal** — trade flowing the other way

 **This is exactly the Chapter 1 lesson paying off.** We can't read Harappan script — but Mesopotamians *could* write, and they wrote about trading with Meluhha. **A readable neighbour becomes a window into an unreadable civilisation.** Beautiful methodological point; use it in Mains.

How did goods move? Bullock carts (terracotta cart models are common finds), pack animals, riverboats, and **seagoing ships** — a Mohenjodaro seal actually depicts a boat with a mast.

8 Society and Religion — Tread Carefully Here

This is where UPSC sets its traps, because the evidence is thin and the temptation to over-read is strong.

What the evidence reasonably supports

| Observation | Reasonable inference |
|--|---|
| Different house sizes, some with more rooms and wells | Some social differentiation existed |
| Citadel vs lower town | Some functional and possibly social separation |
| Burials with pottery, ornaments, occasionally copper mirrors | Belief in an afterlife ; modest grave goods |
| Great Bath, and bathrooms in most houses | Ritual purity / cleanliness mattered greatly |
| Mother Goddess terracotta figurines | Possibly a fertility cult |
| Pashupati seal | Possibly a proto-Shiva / yogic tradition |
| Pipal tree motifs on seals; animal motifs | Nature worship , tree and animal veneration |
| Fire altars at Kalibangan and Lothal | Some form of fire ritual |

✗ What we do NOT have — the elimination checklist

Memorise this list. **Prelims questions are frequently built by asserting one of these.**

| ✗ NOT found | Why it matters |
|--|--|
| No temples | Unlike Mesopotamia's ziggurats and Egypt's temples. No structure is <i>conclusively</i> a temple |
| No palaces | No royal residence identified. We do not know how they were governed |
| No royal names, no king lists | Contrast with Egypt's pharaohs |
| No monumental tombs | Burials are relatively simple. No pyramids |
| No large-scale weapons or fortification for war | Walls exist, but few weapons of war (no swords). Little evidence of organised warfare |
| No horses (reliably), no horse-drawn chariots | Central to Rigvedic culture, absent here |
| No iron | Bronze Age civilisation |
| No coins/money | Trade was barter-based, mediated by standard weights |
| No deciphered script | So no direct knowledge of their beliefs, laws or politics |

 **The "who ruled?" question.** NCERT frames it honestly — we genuinely don't know. Three hypotheses circulate:

1. **A single powerful state** (uniformity of bricks, weights, planning suggests strong central control)
2. **Several city-states** sharing a common culture and standards
3. **A collective/oligarchic or priestly authority** with no single monarch

The correct exam answer acknowledges the uncertainty. Overconfidence here is a wrong answer.

 The uniformity is exactly what you'd get from a **shared standards specification** — which could come from a central authority *or* from a widely-adopted convention among peers. **You cannot distinguish the two from the artefacts alone.** That's the honest state of the field.

The End — Why Did It Collapse?

Around **1900 BCE (~3,900 years ago)** the mature Harappan civilisation unravelled. Cities were abandoned, the script vanished, weights and seals fell out of use, long-distance trade dried up, and people dispersed into smaller village settlements.

What we observe archaeologically

- Drains and street cleaning stop being maintained; rubbish accumulates in streets
- New, flimsier structures built over older planned ones — often encroaching on streets
- Craft specialisation declines; the distinctive pottery, seals and weights disappear
- Writing disappears entirely
- **Population shifts eastward and southward** toward the Ganga-Yamuna region and Gujarat

The competing theories

| Theory | The argument | The problem with it |
|---|--|--|
|  River change / drying ★ (strongest today) | The Ghaggar-Hakra system dried up or shifted course. Over half of all Harappan sites lay along it. The Indus also changed course and flooded | Best supported by geology and site distribution; explains the <i>scale</i> of abandonment |
|  Climate change ★ | Weakening of the monsoon around 2200–1900 BCE (a well-documented global arid event) → agriculture fails → surplus collapses → cities can't be fed | Strong palaeoclimate support; probably works <i>together</i> with the river theory |
|  Deforestation & environmental degradation | Baking millions of bricks and smelting copper consumed enormous fuel wood → deforestation → soil erosion, flooding, reduced rainfall | Plausible contributory factor |
|  Overgrazing / soil exhaustion | Land degraded by intensive use | Contributory |
|  Floods & tectonic shifts | Repeated devastating floods at Mohenjodaro; tectonic uplift damming rivers | Evidence of flooding is real but localised |
|  Collapse of trade | Mesopotamian demand fell; the trade economy that sustained the cities failed | Contributory, not sufficient alone |
|  "Aryan invasion"
 | Proposed by Mortimer Wheeler from some unburied skeletons at Mohenjodaro | Largely REJECTED. The skeletons are from different periods and show no battle wounds; there is no destruction layer, no burnt cities, no mass graves. Do not present this as the accepted view |

★ **The mainstream consensus today: a combination** — primarily **climatic drying and hydrological change** (the Ghaggar-Hakra failing), compounded by environmental degradation and collapsing trade. **Gradual decline and de-urbanisation, not a single catastrophic event or invasion.**

 **The exam-safe answer:** "A multi-causal, gradual de-urbanisation, in which climatic aridification and river-course changes were probably decisive; the invasion hypothesis is no longer accepted." That sentence is both true and well-calibrated.

Late Harappan and continuity

The civilisation didn't vanish — it **transformed**. The **Late Harappan** phase (~1900–1300 BCE) shows smaller, rural settlements continuing, with cultural continuity in pottery, some crafts and agricultural practice. Many Harappan traits (weights, bead-making, cotton, urban water management, possibly some religious motifs) resurface in later Indian culture.

 **A comparison worth holding:** Egypt and Mesopotamia recovered and continued as literate urban societies. **India's urbanism disappeared for over a thousand years** — until the "Second Urbanisation" of the Ganga valley around 600 BCE (Chapter 5). That millennium-long gap is a genuinely distinctive feature of Indian history.

Prelims — the highest-yield checklist in the whole book

| Topic | The exact thing to know |
|------------------------------------|---|
| Site-feature matching ★★ ★ | The #1 format. Master the site table |
| Extremities | Manda (N), Daimabad (S), Alamgirpur (E), Sutkagendor (W) |
| What they did NOT have ★★ ★ | No iron, no horses (reliably), no temples, no palaces, no coins, no royal names |
| Great Bath | Mohenjodaro, citadel, gypsum mortar + bitumen |
| Dockyard | Lothal |
| Ploughed field | Kalibangan (furrows at right angles). Plough model = Banawali |
| No citadel | Chanhudaro |
| Three-part city + signboard | Dholavira (also UNESCO 2021) |
| Fire altars | Kalibangan and Lothal |
| Meluhha / Magan / Dilmun | Indus / Oman / Bahrain |
| Weights | Chert, cubical, binary (16 is key) |
| Faience | An artificial material |
| Artefacts + material | Dancing Girl (bronze), Priest-King (steatite), Male Torso (red sandstone), Pashupati seal (steatite) |
| Script | ~375–400 signs, right-to-left, boustrophedon , undeciphered |
| Excavators | Harappa = Daya Ram Sahni (1921) ; Mohenjodaro = R. D. Banerji (1922) ; announced by John Marshall |
| Decline | Multi-causal; invasion theory rejected |

The three classic traps:

1. "Harappans used horse-drawn chariots in warfare" → **FALSE** (asked in 2013)
2. "Harappans possessed great palaces and temples" → **FALSE** (same question)
3. "Harappans used iron tools" → **FALSE** (Bronze Age)

Mains — how this actually appears

- "Discuss the salient features of Harappan town planning and what they reveal about the nature of political authority."
- "The Harappan civilisation's approach to water management holds lessons for contemporary India. Comment." → **Dholavira's reservoirs, Lothal's dockyard, universal drainage, wells.** A genuinely strong, specific answer.
- **Art & Culture:** seals, the Dancing Girl, terracotta figurines, bead technology, faience.
- **Methodology:** "How does the undeciphered nature of the Harappan script constrain our understanding?" — use the Meluhha/Mesopotamia workaround.

Current-affairs linkages worth knowing

- **Dholavira — UNESCO World Heritage Site (2021)**, India's 40th
- **Rakhigarhi ancient DNA study (2019)** — found no significant Steppe ancestry in the sampled Harappan individual, feeding the Aryan migration debate. Handle neutrally and factually

- Harappan sites and **ASI conservation** issues; Mohenjodaro's deterioration
- **Water harvesting** policy discussions routinely cite Dholavira

Glossary

| Term | Meaning |
|--|---|
| Citadel | The smaller, higher, walled western part of a Harappan city |
| Lower Town | The larger, lower, eastern residential and craft area |
| Great Bath | Large watertight public tank on Mohenjodaro's citadel |
| Faience | Artificial material of ground sand/quartz, glazed and fired |
| Carnelian | Reddish semi-precious stone used for beads |
| Steatite | Soapstone — soft to carve, hardens on firing; used for seals |
| Bitumen | Natural tar; waterproofing agent in the Great Bath |
| Gypsum mortar | Binding material used with bricks for waterproofing |
| Boustrophedon | Writing that alternates direction line by line, "as the ox ploughs" |
| Meluhha | Mesopotamian name for the Harappan region |
| Magan / Dilmun | Oman / Bahrain |
| Sealing | The clay impression left by pressing a seal |
| Chert | Hard stone used for weights and blades |
| Chalcolithic | Copper-Stone Age |
| Proto-history | Period with writing that we cannot read |
| Lost-wax casting (<i>cire perdue</i>) | Bronze-casting technique used for the Dancing Girl |

90-Second Recap

1. **~2600–1900 BCE**, the largest Bronze Age civilisation by area (~1.5 million sq km), contemporary with Egypt and Mesopotamia.
2. **Discovered 1921–22:** Harappa by **Daya Ram Sahni**, Mohenjodaro by **R. D. Banerji**; announced by **John Marshall**.
3. **City plan:** citadel (west, higher, walled) + lower town (east, larger, grid streets).
4. **Drainage is the signature achievement** — covered, sloped, universal, with manholes and soak pits — and **drains were planned before houses**, implying a real civic authority.
5. **Star structures:** Great Bath (Mohenjodaro), dockyard (Lothal), three-part city + signboard + reservoirs (Dholavira), ploughed field + fire altars (Kalibangan), no citadel (Chanhudaro).
6. **Extremities:** Manda (N), Daimabad (S), Alamgirpur (E), Sutkagendor (W); Shortughai in Afghanistan.
7. **Crops:** wheat, barley, pulses, sesame, mustard, **cotton** (world's first); rice at Lothal/Rangpur.
8. **Technology:** bronze, gold, silver, **faience** (man-made), bead-making, shell work, cotton textiles. **NO IRON.**
9. **Seals** of steatite with animal motifs; script of ~375–400 signs, right-to-left, boustrophedon, **undeciphered**.
10. **Weights** of chert, cubical, in a **binary** sequence with 16 as a key unit — standardised across the whole civilisation.

11. **Trade** with **Meluhha/Magan/Dilmun** (Indus/Oman/Bahrain); raw materials sourced from Afghanistan (lapis, tin), Rajasthan (copper), Karnataka (gold), Gujarat (carnelian, shell).
 12. **No temples, palaces, royal names, monumental tombs, or clear evidence of warfare** — governance remains unknown.
 13. **Decline ~1900 BCE:** multi-causal — river drying (Ghaggar-Hakra), climate aridification, environmental degradation, trade collapse. **Invasion theory rejected.**
 14. **Urbanism then disappeared from India for ~1,000 years** until the Ganga valley's Second Urbanisation.
-

Test Yourself

Prelims style

1. Which of the following characterises the people of the Indus Civilisation?
 1. They possessed great palaces and temples.
 2. They worshipped both male and female deities.
 3. They employed horse-drawn chariots in warfare. → *Only 2. (This is UPSC 2013, almost verbatim.)*
2. Consider the following statements:
 1. Kalibangan has yielded evidence of a ploughed field.
 2. Banawali has yielded a terracotta model of a plough.
 3. Chanhudaro is the only major Harappan city without a citadel. → *All three are correct.*
3. In Mesopotamian records, "Meluhha" most probably refers to — → *The Harappan/Indus region.* (Magan = Oman, Dilmun = Bahrain.)
4. Consider the following statements about Dholavira:
 1. It is divided into three parts, unlike the usual two.
 2. It is famous for a large inscription of ten Harappan signs.
 3. It was inscribed as a UNESCO World Heritage Site in 2021. → *All three.*
5. Which of the following was NOT known to the Harappans? → *Iron.* (They knew copper, bronze, gold, silver, faience.)

Mains style

6. *"Harappan town planning reveals more about their civic organisation than their political system."* Discuss. **(150 words)**
7. Discuss the water management practices of the Harappan civilisation and their relevance to contemporary India. **(250 words)**
8. Evaluate the various explanations advanced for the decline of the Harappan civilisation. **(250 words)**

Self-check (out loud, no notes):

- What is in the citadel, and what is in the lower town?
 - Why does "drains were built before houses" matter so much?
 - Name five sites and one unique feature of each.
 - Give three reasons the script remains undeciphered.
 - List five things the Harappans did **not** have.
 - Why is the invasion theory rejected?
-

